

Malaviya National Institute of Technology Jaipur

Department of Computer Science and Engineering



Computer Network And Security Project Report

Session: 2025–2026

50 Shades of support: A Device-Centric Analysis of Android Security Updates

Course Code: 22CPT310

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1 Background

Android has become the most widely used mobile operating system in the world due to its open-source nature, flexibility, and support from multiple device manufacturers. Companies such as Samsung, Xiaomi, Oppo, and Google manufacture Android-based smartphones and tablets for users across different regions and price ranges. With billions of active Android devices globally, mobile security has become a critical concern in the field of cybersecurity. Modern Android devices store large amounts of sensitive information, including banking details, passwords, personal photographs, emails, and business-related data. As a result, attackers continuously attempt to exploit vulnerabilities in Android systems to gain unauthorized access, steal information, or compromise device functionality. These vulnerabilities may exist in the Android operating system, device firmware, chipset drivers, wireless communication modules, or applications. Once discovered, vulnerabilities are assigned Common Vulnerabilities and Exposures (CVE) identifiers and patches are developed to fix them. To improve Android security, the Android Open Source Project (AOSP) releases monthly Android Security Bulletins (ASBs). These bulletins contain information about newly discovered vulnerabilities and their corresponding security patches. Android also uses Security Patch Levels (SPLs) to indicate the latest security updates installed on a device. A recent SPL generally means that the device is protected against a larger number of known vulnerabilities. The Android security update process involves several stages. First, Google releases patches through AOSP. Original Equipment Manufacturers (OEMs) then integrate these patches into their customized Android versions. Mobile carriers may also test and approve updates before they are finally delivered to users through Over-The-Air (OTA) updates. Although this process is designed to improve security, it often becomes slow and inconsistent due to the highly fragmented nature of the Android ecosystem. Unlike centralized operating systems, Android devices vary greatly in terms of hardware specifications, chipset vendors, regional versions, and carrier customizations. OEMs must therefore prepare separate firmware updates for thousands of device variants. This fragmentation leads to irregular update schedules, delayed patch rollouts, and differences in support policies among manufacturers. While some flagship devices receive monthly updates, many low-cost or mid-range devices receive updates quarterly, biannually, or stop receiving updates after a short period. Another major issue is the limited support lifecycle of Android devices. Most smartphones receive security updates for only a few years before reaching their end-of-support (EOS) stage. However, users often continue using these devices because they remain functional. Unsupported devices gradually accumulate vulnerabilities over time, making them easier targets for cyberattacks. Many known vulnerabilities can be exploited remotely and may not require any user interaction, increasing the overall security risk. The research paper “50 Shades of Support: A Device-Centric Analysis of Android Security Updates” studies these issues in detail using a large dataset of Android security updates collected from Samsung, Xiaomi, Oppo, and Google devices between 2014 and 2023. The study analyzes how Android devices are maintained throughout their lifecycle and examines factors affecting update delivery, such as support type, geographical region, device category, and carrier association. The research reveals that Android devices do not follow a single uniform support policy. Devices may transition through monthly, quarterly, and biannual support schedules during their lifetime. The study also shows significant differences between OEMs regarding support duration, update frequency, and release delays. Google Pixel devices generally receive regular monthly updates with minimal delay, whereas other OEMs may provide fewer updates with longer delays depending on device type and region. This project focuses on understanding Android security update behavior and the risks associated with delayed or missing updates. By studying the lifecycle of Android devices and the challenges of patch distribution, the project highlights the importance of timely security maintenance in protecting users against emerging cyber threats. The analysis also helps in understanding how better update policies and support mechanisms can improve the overall security of the Android ecosystem.

2 Introduction

Android is the most widely used mobile operating system in the world, powering billions of smart-phones, tablets, and smart devices. Its open-source architecture and support from multiple Original Equipment Manufacturers (OEMs) such as Samsung, Xiaomi, Oppo, and Google have contributed to its rapid growth and global adoption. However, the diversity of devices and manufacturers in the Android ecosystem has also introduced significant cybersecurity challenges, particularly in the area of security updates and vulnerability management. Mobile devices today store sensitive information including personal data, banking credentials, business communications, and authentication details. As cyber threats continue to evolve, attackers actively exploit vulnerabilities present in Android systems, applications, chipsets, and firmware components. To address these threats, the Android Open Source Project (AOSP) releases monthly Android Security Bulletins containing patches for newly discovered vulnerabilities. These updates are identified through Security Patch Levels (SPLs), which indicate the latest security fixes installed on a device. Despite the availability of regular patches, the delivery of security updates across Android devices remains inconsistent. Unlike centralized ecosystems, Android updates must pass through multiple stages involving OEM customization, carrier testing, and regional deployment before reaching end users. Due to device fragmentation, varying hardware configurations, and manufacturer-specific policies, many Android devices experience delayed updates or stop receiving support entirely after a limited period. Consequently, users often continue operating devices with outdated software, leaving them vulnerable to known cyberattacks. This project focuses on analyzing the Android security update ecosystem using a device-centric approach based on the research paper “50 Shades of Support: A Device-Centric Analysis of Android Security Updates.” The study examines the lifecycle of Android devices, factors affecting update distribution, and the security risks associated with unsupported devices. It also evaluates the impact of support type, geographical region, carrier association, and device category on the timeliness and availability of updates. The objective of this project is to understand the challenges in Android security maintenance and highlight the importance of timely patch management in protecting users against emerging vulnerabilities. Through this analysis, the project aims to provide insights into improving Android security practices and enhancing the overall cybersecurity of mobile ecosystems.

3 Android Security Update Architecture

3.1 Android Security Update Process

The Android security update process involves multiple stages and stakeholders before a security patch reaches end users. The process begins when vulnerabilities are identified in the Android operating system, device firmware, kernel components, or hardware drivers. These vulnerabilities are assigned Common Vulnerabilities and Exposures (CVE) identifiers and are analyzed by security researchers and the Android Security Team.

After vulnerabilities are identified, patches are developed and released through the Android Open Source Project (AOSP). Google publishes monthly Android Security Bulletins (ASBs) containing detailed information regarding affected components, severity levels, and corresponding fixes. These bulletins form the foundation of Android’s security maintenance mechanism.

Original Equipment Manufacturers (OEMs) such as Samsung, Xiaomi, Oppo, and others then integrate these patches into their customized Android firmware. Since OEMs modify Android using custom user interfaces, drivers, and proprietary applications, the integration process requires additional testing and optimization for each device model. Mobile carriers may also verify updates for network compatibility before approving deployment.

Finally, the updates are distributed to users through Over-The-Air (OTA) updates. Depending on the manufacturer and device category, updates may be released monthly, quarterly, or biannually. However, due to Android fragmentation and varying OEM policies, the update delivery process is often delayed and inconsistent across devices and regions.

3.2 Android Security Bulletins

Android Security Bulletins (ASBs) are monthly reports published by Google that provide details about newly discovered vulnerabilities and their corresponding security patches. These bulletins are an essential part of Android's cybersecurity framework and help OEMs maintain device security against emerging threats.

Each bulletin includes information regarding:

Vulnerability identifiers (CVEs) Severity levels Affected Android components Patch availability Security Patch Level (SPL) dates

The vulnerabilities listed in ASBs may affect different parts of the Android ecosystem, including the Android framework, kernel, media framework, system libraries, Qualcomm chipsets, and wireless communication modules. Vulnerabilities are categorized based on severity levels such as Critical, High, Moderate, and Low depending on their potential impact.

Critical vulnerabilities are particularly dangerous because they may allow remote code execution without requiring user interaction. Attackers can exploit such vulnerabilities to gain unauthorized access to a device, steal sensitive data, or compromise system functionality.

Android Security Bulletins also help OEMs prioritize patch deployment and maintain transparency regarding Android security risks. Devices receiving regular security bulletins are generally more protected against newly discovered cyber threats.

3.3 Security Patch Levels (SPL)

Security Patch Level (SPL) is a metric represented by a date. The date indicates when the last set of security patches were deployed on the device. SPL can help evaluate how secure a device is against threats.

A SPL is typically reported for Android devices as follows:

YYYY-MM-DD

For instance,

2025-04-05

An older SPL indicates that there are fewer security patches applied to the device. Two kinds of SPL are issued by Google each month:

01 security patch level 05 security patch level

01 security patch level includes partial security patches while 05 SPL includes security patches related to the Android framework and hardware vendor, such as Qualcomm drivers and firmware.

The significance of SPL is best illustrated by the fact that the older the SPL, the greater the vulnerability to risks with public exploits available. Typically, hackers target devices with the older SPL because no protection exists against these risks.

SPLs are released based on the OEM support policy. Flagship phones and tablets receive monthly SPL updates while others receive inconsistent updates.

3.4 Android Device Lifecycle

An Android device also has a specific support lifecycle, meaning that some software and security updates are provided within the set timeframe starting from the moment when the device is introduced into the market. Therefore, a support lifecycle of an Android device begins on the day of its introduction and ends after going through several different stages that comprise several update cycles, resulting in an EOS status of the particular gadget.

At first, most of the devices receive security and other updates every month during several years following the day of introduction. It means that at that stage the device receives high attention and all discovered problems and vulnerabilities are fixed.

The following phase is characterized by an increasing interval between the updates. Some devices start receiving only two updates per year, whereas cheaper devices may not receive updates once in

three months. Longer intervals between the updates mean that some vulnerabilities are not fixed for an extended period of time.

Finally, an EOS stage arrives, when a device does not receive any more updates. Nevertheless, the device continues functioning. Yet, due to an increasing amount of vulnerabilities that emerge every year, the risk of a threat grows significantly.

4 Experimental Setup

4.1 Hardware

- Laptop: MacBook Pro M4 Pro
- RAM: 24 GB
- OS: macOS Tahoe 26.3.1

4.2 Software

- Jupyter Notebook

4.3 Datasets

- https://nvd.nist.gov/vuln/data-feeds#JSON/_FEED

5 Implementation

1. Base Project

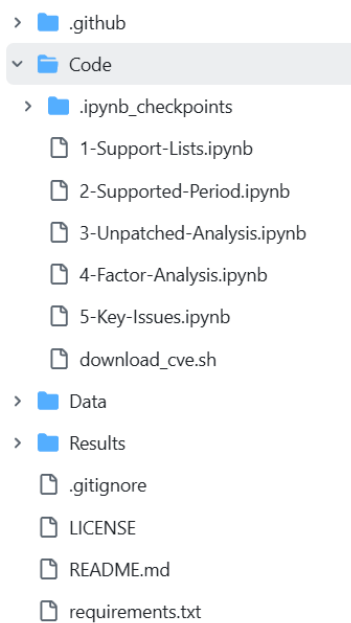


Figure 1: Project Structure

6 Results

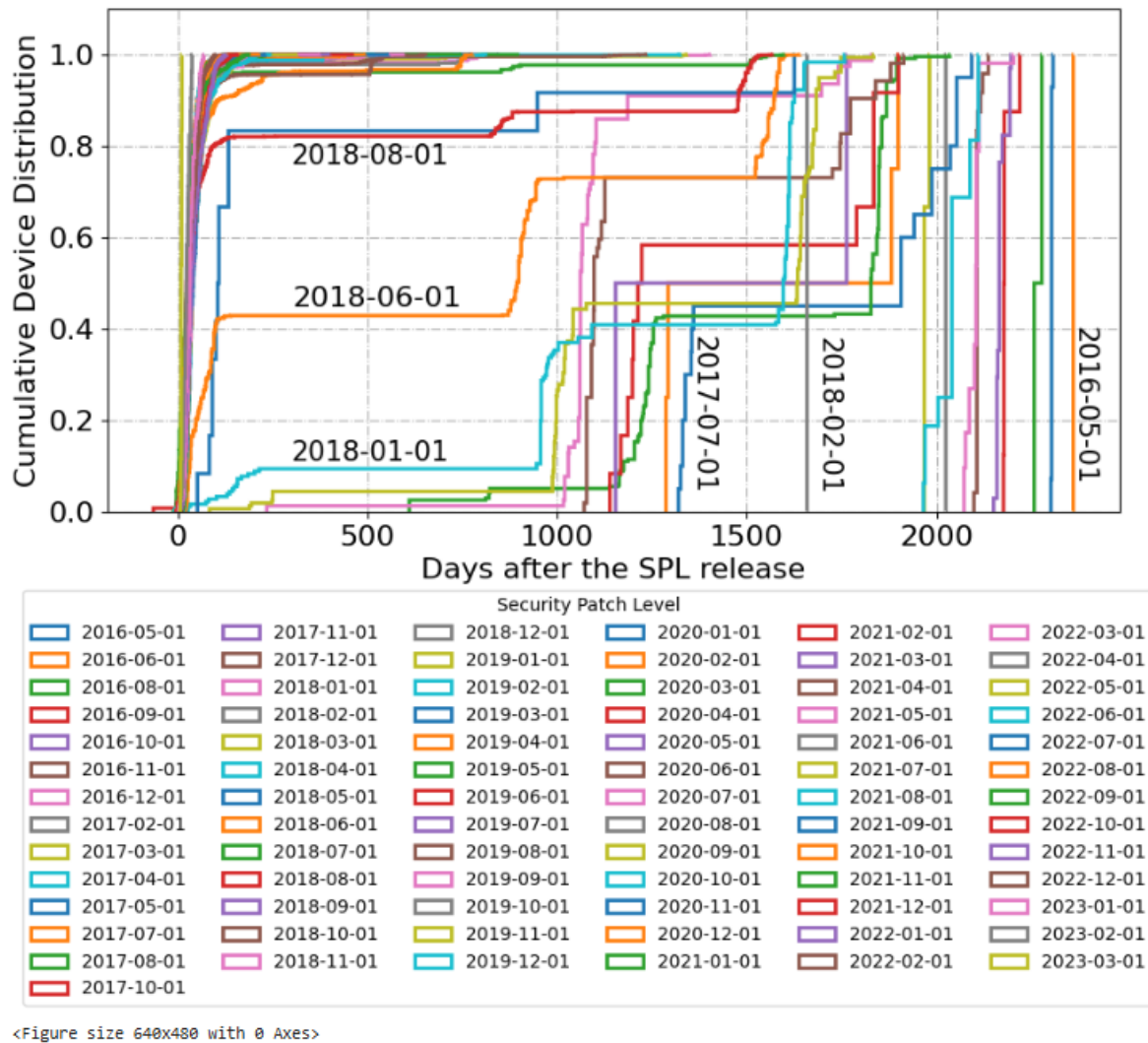


Figure 2:

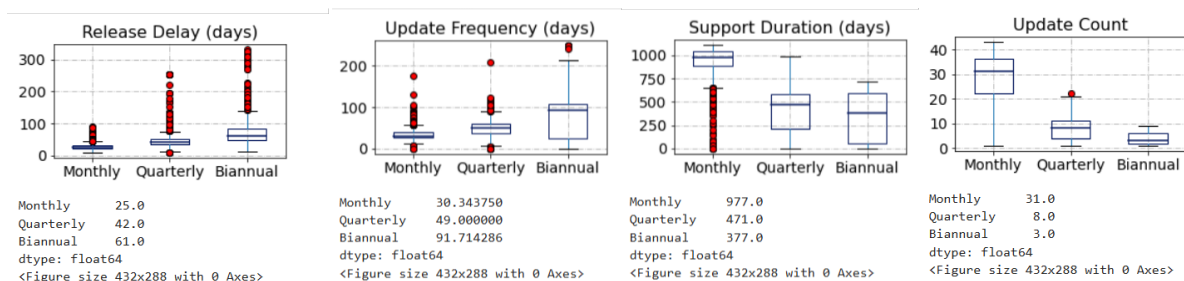


Figure 3:

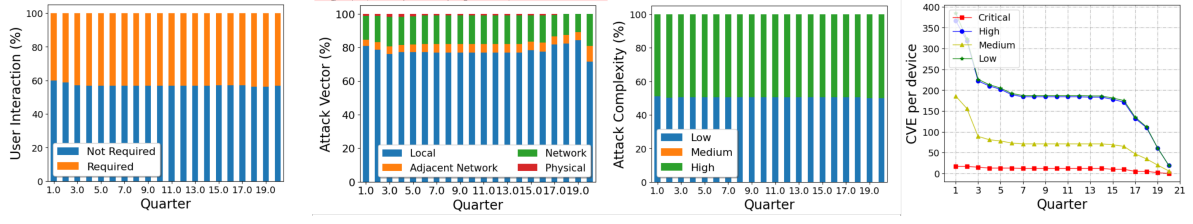


Figure 4:

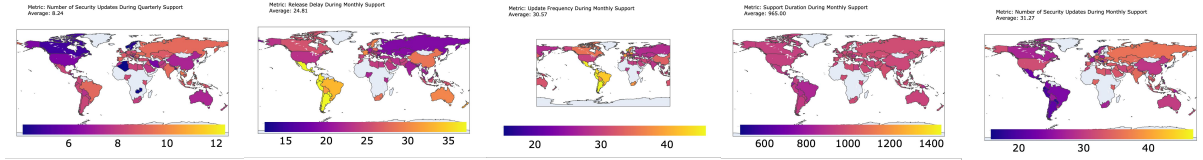


Figure 5:

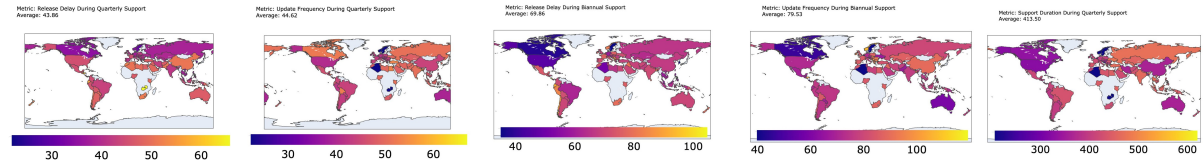


Figure 6:

7 Novel Work

7.1 Safe Usage Window (SUW) Estimation

One of the important updates made within this project involves the introduction of the Safe Usage Window (SUW) metric, which characterizes unsupported Android devices. The initial project was based mostly on determining and calculating the amount of vulnerabilities which persist in a device past its EOL date. However, no metrics were provided regarding the period during which this type of devices can be used while having a reasonably safe level of accumulated vulnerabilities.

In order to fill this gap, an estimation model for calculating Safe Usage Windows for unsupported devices was devised. It evaluates the approximate duration for which a device can operate having a relatively safe level of threat exposition before it becomes critically insecure because of the growing number of vulnerabilities. This evaluation is based on the characteristics of post-EOL vulnerabilities, like their quantity and degree of exploitability.

This improvement makes the project more practical and useful since it turns it into a security lifecycle evaluation framework rather than simple research of vulnerabilities accumulation. With the Safe Usage Window, users can calculate how soon they should replace their devices, create proper enterprise mobile security policy, and evaluate other aspects related to the prolonged use of unsupported smartphones.

7.2 Dynamic Device Risk Score (DSRS)

Another innovative element of the DSRS scheme introduced in this project deserves mention. At the first stage of development, security risks were evaluated based on the number of vulnerabilities connected to

unsupported devices. However, not all vulnerabilities have equal chances of being used in attacks.

To resolve this issue, a new exploitability-oriented DSRS scoring scheme was developed. The following vulnerability parameters are considered within the DSRS scoring scheme:

- Severity
- Attack vector
- Attack complexity
- Necessity for user interaction

Thus, a vulnerability with a high score can be:

- Remotely exploited
- Severe in nature
- Simple to exploit
- Easily exploited without user interaction

The score will be assigned to each device, which will indicate the extent of cybersecurity risks it faces after EOL. This innovation greatly increases the value of the project's contribution to cybersecurity since it introduces a much more practical evaluation of the risks.

Furthermore, the DSRS framework provides for:

- Vendor comparison
- Identification of vulnerable devices
- Ranking of unsupported devices from a cybersecurity risk perspective

7.3 Post-EOL Exploitability Exposure Analysis

The third major development introduced in this study is Post-EOL Exploitability Exposure Analysis. As was previously noted, the main finding from the first project was that despite the fact that the devices stopped getting updates on security, they continued having vulnerabilities; however, this project did not analyze the extent to which these vulnerabilities were exploitable.

To enrich the security implications of the project, the exploitability characteristics of each identified vulnerability were evaluated. Specifically, for each of them the following features were considered:

- Severity level
- Attack vector
- Attack complexity
- User interaction

Thus, in this study, it became possible to identify vulnerabilities that were

- Remotely exploitable vulnerabilities
- Extremely critical vulnerabilities
- Vulnerabilities requiring no user interaction

In other words, unlike the previous project, now it does not only provide the answer to the question of

”How many vulnerabilities are there after EOL?”

But also answers the question of

”How exploitable are those vulnerabilities?”

This makes the project much more relevant for cybersecurity practices.

7.4 Images

				SafeUsageWindow_months			
	id	vendor	DSRS		id	vendor	
11680	SM-G780G/CAU	samsung	2243.35	14174	SM-G970F/MWD	samsung	36
10524	SM-X706B/ILO	samsung	2243.35	2512	SM-G960F/XME	samsung	36
10530	SM-X706B/PAK	samsung	2243.35	7537	SM-G715U1/USC	samsung	36
10529	SM-X706B/MWD	samsung	2243.35	6119	SM-A105FN/SFR	samsung	36
10528	SM-X706B/MM1	samsung	2243.35	12061	SM-G955W/KD0	samsung	0
10527	SM-X706B/MID	samsung	2243.35	9415	SM-S901E/CGU	samsung	36
10526	SM-X706B/KSA	samsung	2243.35	9197	SM-G990E/XFE	samsung	36
10525	SM-X706B/INU	samsung	2243.35	13680	SM-A510F/THL	samsung	0
10523	SM-X706B/EGY	samsung	2243.35	18505	SM-N9810/BRI	samsung	36
10481	SM-T976B/XEF	samsung	2243.35	1507	SM-G965F/XTC	samsung	36
10522	SM-X706B/ECT	samsung	2243.35	21717	ido_xhdpi_global	xiaomi	0
10521	SM-X706B/CHO	samsung	2243.35	8693	SM-A235M/IUS	samsung	36
10520	SM-X706B/CAU	samsung	2243.35	19407	SM-G950F/BOG	samsung	0
10519	SM-X706B/BVO	samsung	2243.35	12855	SM-G930F/ECO	samsung	0
10518	SM-X706B/BRI	samsung	2243.35	6931	SM-A405FN/ITV	samsung	36
10517	SM-X706B/AFR	samsung	2243.35	12196	SM-J710FN/INS	samsung	0
10531	SM-X706B/SEK	samsung	2243.35	7592	SM-F700F/BGL	samsung	36
10532	SM-X706B/SER	samsung	2243.35	203	SM-J330F/TPL	samsung	0
10533	SM-X706B/SIN	samsung	2243.35	8432	SM-A217F/EVR	samsung	36
10534	SM-X706B/SKZ	samsung	2243.35	11330	SM-P619/UY0	samsung	36
10535	SM-X706B/SLK	samsung	2243.35	10068	SM-G998B/XEH	samsung	36
10536	SM-X706B/STH	samsung	2243.35	14139	SM-G970F/CYV	samsung	36
10537	SM-X706B/TEL	samsung	2243.35	18872	SM-X205/KSA	samsung	36
10538	SM-X706B/TGY	samsung	2243.35	2952	SM-A115F/CAC	samsung	36
10539	SM-X706B/THL	samsung	2243.35	7248	SM-F926B/PRT	samsung	36
10540	SM-X706B/TUN	samsung	2243.35	2886	SM-A107M/TPA	samsung	36
10541	SM-X706B/UP0	samsung	2243.35	16466	SM-G950U/USC	samsung	0
10542	SM-X706B/XFA	samsung	2243.35	9708	SM-S906E/ECT	samsung	36
10543	SM-X706B/XFE	samsung	2243.35	7032	SM-A505G/SAM	samsung	36
10544	SM-X706B/XFV	samsung	2243.35	21854	munch_global	xiaomi	36

Figure 7:

8 Future scope/Improvements

8.1 Predictive Security Risk Scoring System

One of the issues that arise in the context of the Android ecosystem is the lack of ability for the user to correctly evaluate the level of risks associated with using an outdated device. While Android devices show SPLs, they do not give any indications regarding the level of vulnerability of the device in relation to cyber threats existing at the moment. Users operate their devices with delayed or missing updates, unaware of the extent of the risk involved.

In order to solve this problem, a Predictive Security Risk Scoring System is suggested as a novel contribution in the current research. It allows for evaluation of Android devices and scoring them with a numeric indicator of risk depending on several factors related to security of the device such as:

- Current Security Patch Level (SPL)
- Number of unpatched CVEs

- Severity of vulnerabilities
- Age of the device
- OEM update policies
- Previous update delays

The suggested model will continually analyze Android Security Bulletins and databases of vulnerability and predict the risk value for the particular device. The higher the SPL value and the more unpatched CVEs there are, the higher the numeric score will be.

8.2 AI-Based Security Update Delay Prediction

Patch rollouts are not predictable in the Android OS since while some devices may be getting their security updates every month promptly, other devices are having a haphazard patch rollout schedule.

As a novel concept, this research proposes an idea of developing an AI-based Security Update Delay Prediction model that can help predict how the security updates would behave in the future according to certain patterns seen before. Predictive patterns may include:

- History of update releases for OEMs
- Type of the device
- Location (geography)
- Affiliation to a particular carrier
- Lifespan of the Android OS version
- Delay times of past update releases

Machine learning techniques are used in this model to predict delays of future security updates in advance. Devices with longer predicted delay intervals are high-risk devices.

AI-based prediction models for security update delays could help organizations manage device management and compare security update delay performance of different OEMs.

8.3 Automated Vulnerability Exposure Monitoring Dashboard

A significant problem identified in the Android ecosystem is the lack of real-time visibility regarding device vulnerability exposure. Most Android users are unaware of the number and severity of vulnerabilities affecting their devices, especially after reaching end-of-support status.

To solve this problem, this project proposes an Automated Vulnerability Exposure Monitoring Dashboard as a novel work. The dashboard continuously monitors Android devices using publicly available security feeds including:

- Android Security Bulletins
- National Vulnerability Database (NVD)
- Common Vulnerabilities and Exposures (CVE) databases

The system maps vulnerabilities to devices using Android version information, chipset details, and Security Patch Levels. It then generates real-time security analytics such as:

- Total unpatched vulnerabilities

- Critical CVE count
- Device support status
- Security health indicators
- Vulnerability severity distribution

The dashboard also provides visual alerts for high-risk devices and displays support expiration timelines. Organizations can use the dashboard for centralized Android device security monitoring and risk assessment.

This novel system improves security awareness and enables proactive identification of vulnerable Android devices before exploitation occurs.